1, Team Number

- 007

2. Team Name

- Double-O-7

3. Team Members

- Pierce Le
- Korah Coulter
- Alexander Morgan
- Payton Dozois
- Jaron King

4. Application Name

- Trackademic

5. Application Description

Trackademic is a web application designed to help students manage their academic workload throughout the semester. Users can upload a course syllabus in a PDF format, and the application automatically parses the document to extract key dates, assignments, and exams, populating a visual calendar with those events. Each week is then analyzed and assigned a difficulty rating based on the density and weight of upcoming deadlines, giving students a clear picture of when their heaviest workloads are approaching. With a personalized dashboard, students can plan ahead, avoid last-minute cramming, and stay on top of their coursework across multiple classes.

Each week on the calendar is analyzed and assigned a difficulty rating based on the number and weight of upcoming deadlines, visually indicated through a color coded system ranging from light to heavy workload weeks. This allows students to identify crunch periods well in advance and plan their time accordingly, reducing last minute stress and improving overall academic performance. Users can also manually add, edit, or delete events to account for any information the parser may have missed or any changes made by their instructor throughout the semester. Lastly another key feature is that the application can help suggest when to start projects

6. Audience

The primary target audience for Trackademic is college and university students, particularly those juggling multiple courses at once who struggle to track deadlines, exams, and assignments scattered across different syllabi. These users are comfortable with basic web

applications but are not necessarily technically advanced, so the application is designed to be intuitive and require minimal setup to deliver value immediately. We look to serve all students but particularly students who are academically inclined and want to use their time studying rather than planning.

7. Vision Statement

- The one stop shop to help students visualize and manage their workloads before their workloads manage them.

8. Version Control

- https://github.com/pifyy/syllabus-planner.git

9. Development Methodology

- The development methodology we are using will be a mix of AGILE and SCRUM, we will have weekly standup meetings where stories will be updated, roadblocks will be discussed. We plan on having two standup meetings per week and will dedicate a SCRUM master to run these meetings.

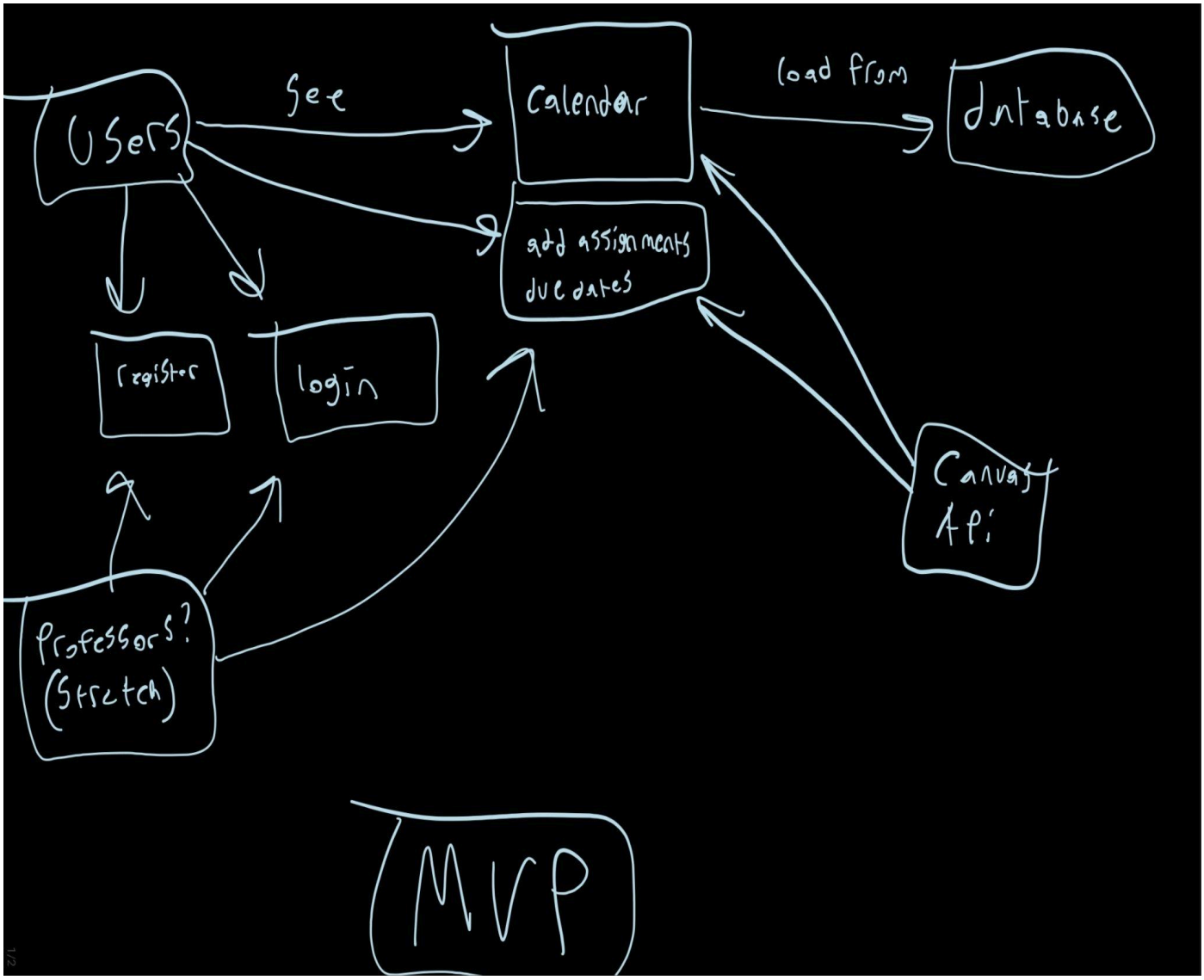
10. Communication Plan

- We plan on communicating to the entire team via Discord. Here is the link to our server
- https://discord.gg/p9A8apG5

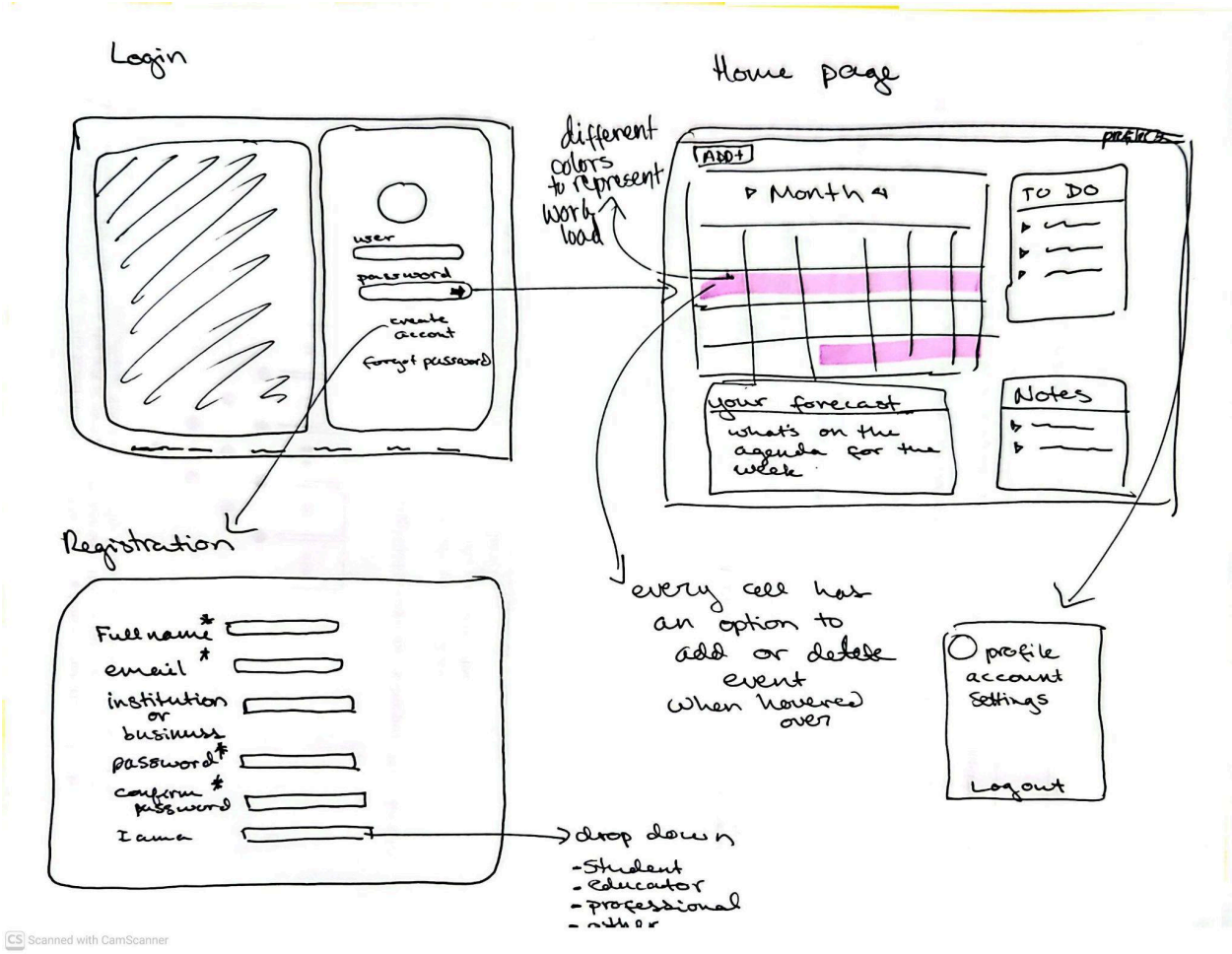
11. Meeting plan

- Team Meeting, Mondays Wednesdays after class, with additional zoom meetings as needed.
- Meeting with TA, TBD as Zach mentions it to us, we have filled out everything on our part though.

12. Use Case Diagram



13. Wireframes



Scanned with CamScanner

Extra Credit

- External API Change:
  - Severity: Medium
  - Challenge: One of the challenges we might face during development is if one or multiple of the APIs used in the application changes. This could break core functionality such as pdf parsing or calendar re-rendering mid project.
  - Approach: To deal with this the team will identify backup dependencies to add to the package.json folder early on.
- Creating too much scope:
  - Severity: Light
  - Challenge: With only four weeks to develop this product, if we are to add too many features we might not be able to create a product at all resulting in an incomplete product by the time of the deadline.
  - Approach: we are going to well define the scope of the product and get the foundation of the web application done, additional features will be added once everything foundational is functional and correct.
- LLM parsing uploaded syllabi:
  - Severity: Medium

    - Challenge: We would need to find a balance between an LLM that accurately and meaningfully parses information from uploaded documents that also isn't too operationally expensive.
    - Approach: Accurately assess what we need, test to ensure that there is not a heavy wait time, and be ready to adapt if one approach does not work.
- Data Privacy and Security
    - Severity: High
    - Challenge: Since users are uploading academic information to their accounts, there is a chance that improper implementation of database queries can return sensitive data to unauthorized users.
    - Approach: To mitigate this, the team will ensure all routes that access user data are protected by authentication middleware, all passwords are hashed and database queries are parameterized to prevent SQL injection.
- Communication, time management, and division of labor
    - Severity: Medium
    - Challenge: anytime individuals are asked to operate in a group setting, there is the risk of operating out of sync. It is important that work is divided evenly and assigned work is completed in timely manners.
    - Approach: Maintain communication and check in with each other. Privately talk to team members who may feel challenged or are falling behind.